

Practical 13 - GLMs for grouped binary and count outcomes - Stata

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `medpar.csv`

In this practical we will explore models for a count variable and look at the two approaches described in the lecture for handling overdispersion. We will use publicly available data on length of hospital stay from Arizona in the US.

In our analyses we will focus on the following variables.

Variable	Description
<code>los</code>	Length of hospital stay, in days
<code>age</code>	Age group (factor variable)
<code>type1</code>	Binary variable indicating elective admission
<code>type2</code>	Binary variable indicating urgent admission
<code>type3</code>	Binary variable indicating emergency admission

Note that admission type is here described using three binary indicator variables, rather than as a 3-level categorical variable.

Aims

1. Understand how to fit models to count data using the `glm` command.
2. Understand how to compare such models.
3. Understand the concept of overdispersion, and how to identify it.

Question 1

Explore the length of stay variable. How is it distributed? What are the minimum and maximum values in this dataset?

Construct a 95% confidence interval for the mean length of stay using the standard approach taught in the Analytical Techniques course.

SOLUTIONS

```
. summ los, d
```

los				

Percentiles		Smallest		
1%	1	1		
5%	1	1		
10%	2	1	Obs	1,495
25%	4	1	Sum of wgt.	1,495
50%	8	Largest	Mean	9.854181
			Std. dev.	8.832906
75%	13	70		
90%	19	74	Variance	78.02022
95%	23	91	Skewness	3.651526
99%	46	116	Kurtosis	29.20186

```
. ci means los
```

Variable	Obs	Mean	Std. err.	[95% conf. interval]	
-----+-----					
los	1,495	9.854181	.2284457	9.406072	10.30229

The 95% CI extends from 9.406 to 10.302. This CI is valid provided the variable is normally distributed. Even if the variable is not normally distributed the coverage of the CI will be close to 95% if the sample size is sufficiently large. This is a consequence of the central limit theorem. Here the coverage should be reasonable since the sample size is quite large.

Question 2

Use the `glm` function to fit a Poisson model for length of stay, with no covariates.

What does the constant coefficient in this model represent? Construct a 95% confidence interval for the mean length of stay.

What can you infer from the reported deviance statistic?

SOLUTIONS

```
. glm los, fam(poisson) link(log) nolog
```

Generalized linear models		Number of obs	=	1,495
Optimization	: ML	Residual df	=	1,494
		Scale parameter	=	1
Deviance	= 8901.134077	(1/df) Deviance	=	5.957921
Pearson	= 11828.70662	(1/df) Pearson	=	7.917474
Variance function:	V(u) = u			[Poisson]
Link function	: g(u) = ln(u)			[Log]
		AIC	=	9.778116
Log likelihood	= -7308.141824	BIC	=	-2019.829

		OIM					
los		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	

_cons		2.287896	.0082389	277.69	0.000	2.271748	2.304044

The residual deviance (8901.1) is substantially larger than the residual degrees of freedom (1494). This strongly suggests overdispersion.

Question 3

Re-run the `glm` with the `robust` option to obtain robust sandwich SEs and CIs.

Compare and contrast the 95% confidence intervals in questions 1, 2 and 3. Which is most appropriate? Which is least appropriate?

SOLUTIONS

```
. glm los, fam(poisson) link(log) nolog robust
```

```
Iteration 0: log pseudolikelihood = -7354.5568
```

```

Iteration 1:  log pseudolikelihood = -7308.2078
Iteration 2:  log pseudolikelihood = -7308.1418
Iteration 3:  log pseudolikelihood = -7308.1418

```

```

Generalized linear models      Number of obs   =      1,495
Optimization      : ML        Residual df      =      1,494
                               Scale parameter =          1
Deviance          =  8901.134077 (1/df) Deviance =   5.957921
Pearson           = 11828.70662   (1/df) Pearson  =   7.917474

```

```

Variance function: V(u) = u      [Poisson]
Link function      : g(u) = ln(u) [Log]

```

```

                               AIC          =    9.778116
Log pseudolikelihood = -7308.141824      BIC          =   -2019.829

```

	los	Coefficient	Robust std. err.	z	P> z	[95% conf. interval]	
	_cons	2.287896	.0231826	98.69	0.000	2.242459	2.333333

The estimated mean is the same using all three approaches. The 95% CI is much narrower when making the Poisson assumption (9.696 to 10.015) than using either of the other two approaches (9.406 to 10.302 and 9.416 to 10.312), which are very similar to each other.

The difference between the 95% CIs in questions 2 and 3 can be attributed to overdispersion in `los`: the deviance statistic is much larger than its d.f. This makes sense given the context; the underlying severity of incoming patients' conditions will vary, leading to more variability than predicted by the Poisson distribution. So, the approach in question 2 is not appropriate.

The approach in question 1 takes no account of the skewness of the distribution of `los` (the 95% CI here is symmetric about the mean on an arithmetic scale), so the approach in question 3 is probably to be preferred (here the 95% CI is symmetric about the mean on a multiplicative scale), but (because of the large sample size and the central limit theorem) the approach in question 1 is also acceptable.

Question 4

We will now explore which (if any) factors are related to length of hospital stay. First fit a Poisson model to assess whether length of hospital stay is related to the type of admission.

Interpret each of the parameter estimates in your model.

SOLUTIONS

Note that we only need to include two of the indicator variables (we cannot include all three). We'll include `type2` and `type3`, so that type 1 becomes the reference level:

```
. glm los i.type2 i.type3, fam(poisson) link(log) eform
```

```
Iteration 0:   log likelihood = -7029.5631
Iteration 1:   log likelihood = -6949.5862
Iteration 2:   log likelihood = -6949.3886
Iteration 3:   log likelihood = -6949.3886
```

Generalized linear models		Number of obs	=	1,495
Optimization	: ML	Residual df	=	1,492
		Scale parameter	=	1
Deviance	= 8183.627612	(1/df) Deviance	=	5.485005
Pearson	= 9396.93324	(1/df) Pearson	=	6.298213

Variance function: $V(u) = u$	[Poisson]
Link function : $g(u) = \ln(u)$	[Log]

	AIC	=	9.300854
Log likelihood = -6949.388592	BIC	=	-2722.716

		OIM					
	los	IRR	Std. Err.	z	P> z	[95% Conf. Interval]	

1.type2		1.267877	.0265014	11.35	0.000	1.216985	1.320897
1.type3		2.065477	.0535019	28.00	0.000	1.963233	2.173046
_cons		8.830688	.0882451	217.98	0.000	8.659413	9.00535

Note: `_cons` estimates baseline incidence rate.

```
. est store A
```

Note that in the above we have used `eform` to get Stata to exponentiate the coefficients for us.

The mean length of stay for elective admissions is 8.83 days (95% CI 8.66 to 9.01 days). Urgent admissions are associated with stays that are on average 26.8% (95% CI 21.7% to 32.1%) longer

than those resulting from elective admissions. Emergency admissions result in stays that are on average 2.06 (95% CI 1.96 to 2.17) times as long as those from elective admissions.

However, note that all of these 95% CI rely on the Poisson assumption being correct and in fact this assumption is likely incorrect due to overdispersion as described above.

Question 5

Add age as factor variable to your model in question 4 and perform a likelihood ratio test of whether age (treated as categorical) is a predictor of length of hospital stay, after adjusting for type of admission. Use the reported deviance to infer whether this is an appropriate test.

SOLUTIONS

```
. glm los i.age i.type2 i.type3, fam(poisson) link(log) nolog eform
```

Generalized linear models		Number of obs	=	1,495
Optimization	: ML	Residual df	=	1,484
		Scale parameter	=	1
Deviance	= 8165.18541	(1/df) Deviance	=	5.502147
Pearson	= 9346.752373	(1/df) Pearson	=	6.298351
Variance function: V(u)	= u	[Poisson]		
Link function	: g(u) = ln(u)	[Log]		
		AIC	=	9.299221
Log likelihood	= -6940.167491	BIC	=	-2682.679

		OIM					
	los	IRR	Std. Err.	z	P> z	[95% Conf. Interval]	

	age						
	2	.7755658	.0956944	-2.06	0.039	.6089643	.9877464
	3	.8176414	.0972513	-1.69	0.091	.6476194	1.0323
	4	.7910486	.0931576	-1.99	0.047	.6280029	.9964251
	5	.7659533	.0901755	-2.26	0.024	.6081218	.9647483
	6	.7951697	.0935329	-1.95	0.051	.6314458	1.001345
	7	.7593596	.0901184	-2.32	0.020	.6017686	.9582205
	8	.7239164	.0878831	-2.66	0.008	.5706278	.9183832
	9	.7316548	.09231	-2.48	0.013	.5713648	.9369124
	1.type2	1.265918	.0265407	11.25	0.000	1.214954	1.319021

1.type3	2.064878	.053619	27.92	0.000	1.962417	2.17269
_cons	11.32913	1.320198	20.83	0.000	9.01581	14.236

Note: _cons estimates baseline incidence rate.

```
. est store B
```

```
. lrtest A B
```

Likelihood-ratio test	LR chi2(8) =	18.44
(Assumption: A nested in B)	Prob > chi2 =	0.0181

The p-value from the likelihood ratio test is 0.018 - we have some evidence that there are differences in (log) mean length of stay between the different age groups after adjusting for admission type.

But the residual deviances in both models are much larger than the residual d.f. As such, the likelihood ratio test may not be reliable here for assessing whether there is an effect of age group conditional on admission type. We can however take the small p-value as evidence that the Poisson model with just admission type as covariate is not correctly specified.

Question 6

Use the `nbreg` command to fit negative binomial models analogous to those you fitted in questions 4 and 5. Use `lrtest` to perform a likelihood ratio test comparing the two negative binomial models. Compare your result with the previous question's findings.

SOLUTIONS

```
. nbreg los i.type2 i.type3, irr
```

Fitting Poisson model:

```
Iteration 0: Log likelihood = -6949.6915
Iteration 1: Log likelihood = -6949.3886
Iteration 2: Log likelihood = -6949.3886
```

Fitting constant-only model:

```
Iteration 0: Log likelihood = -4988.8172
Iteration 1: Log likelihood = -4857.3372
Iteration 2: Log likelihood = -4856.494
```

Iteration 3: Log likelihood = -4856.494

Fitting full model:

Iteration 0: Log likelihood = -4802.8473

Iteration 1: Log likelihood = -4800.2234

Iteration 2: Log likelihood = -4800.2189

Iteration 3: Log likelihood = -4800.2189

Negative binomial regression

Number of obs = 1,495

LR chi2(2) = 112.55

Dispersion: mean

Prob > chi2 = 0.0000

Log likelihood = -4800.2189

Pseudo R2 = 0.0116

los	IRR	Std. err.	z	P> z	[95% conf. interval]	
1.type2	1.267877	.0636685	4.73	0.000	1.149033	1.399012
1.type3	2.065477	.1563594	9.58	0.000	1.780669	2.395838
_cons	8.830688	.196424	97.93	0.000	8.453975	9.224187
/lnalpha	-.8033559	.0443828			-.8903446	-.7163671
alpha	.4478236	.0198757			.4105142	.4885238

Note: Estimates are transformed only in the first equation to incidence-rate ratios.

Note: _cons estimates baseline incidence rate.

LR test of alpha=0: chibar2(01) = 4298.34

Prob >= chibar2 = 0.000

. est store nbA

. nbreg los i.age i.type2 i.type3, irr

Fitting Poisson model:

Iteration 0: Log likelihood = -6940.469

Iteration 1: Log likelihood = -6940.1675

Iteration 2: Log likelihood = -6940.1675

Fitting constant-only model:

Iteration 0: Log likelihood = -4988.8172

Iteration 1: Log likelihood = -4857.3372

Iteration 2: Log likelihood = -4856.494
 Iteration 3: Log likelihood = -4856.494

Fitting full model:

Iteration 0: Log likelihood = -4800.8134
 Iteration 1: Log likelihood = -4797.8059
 Iteration 2: Log likelihood = -4797.7982
 Iteration 3: Log likelihood = -4797.7982

Negative binomial regression	Number of obs = 1,495
	LR chi2(10) = 117.39
Dispersion: mean	Prob > chi2 = 0.0000
Log likelihood = -4797.7982	Pseudo R2 = 0.0121

los	IRR	Std. err.	z	P> z	[95% conf. interval]	
age						
2	.736538	.2296887	-0.98	0.327	.3997134	1.357193
3	.8104465	.245	-0.70	0.487	.44813	1.465699
4	.7720919	.2315234	-0.86	0.388	.4289665	1.389679
5	.734201	.2200139	-1.03	0.303	.4080751	1.32096
6	.7792436	.2333794	-0.83	0.405	.4332542	1.401534
7	.7467241	.2251701	-0.97	0.333	.4135095	1.34845
8	.7017948	.2152102	-1.15	0.248	.3847542	1.28008
9	.7181815	.2273518	-1.05	0.296	.3861649	1.335659
1.type2	1.269589	.0638731	4.74	0.000	1.150374	1.401158
1.type3	2.077296	.157614	9.64	0.000	1.79025	2.410366
_cons	11.61195	3.447394	8.26	0.000	6.489294	20.77842
/lnalpha	-.8069933	.0444229			-.8940606	-.719926
alpha	.4461976	.0198214			.4089916	.4867883

Note: Estimates are transformed only in the first equation to incidence-rate ratios.

Note: _cons estimates baseline incidence rate.

LR test of alpha=0: chibar2(01) = 4284.74 Prob >= chibar2 = 0.000

. est store nbB

. lrtest nbA nbB

Likelihood-ratio test

Assumption: nbA nested within nbB

```
LR chi2(8) =    4.84
Prob > chi2 = 0.7744
```

We now have a p-value 0.77 for testing whether the age groups differ after conditioning on type of admission. This is a completely different result from the Poisson likelihood ratio test comparison from the previous question. The explanation is that the negative binomial model allows for the possibility of overdispersion.

Question 7

Write a short paragraph (suitable for a medical journal) to summarise your findings concerning the effects of the type of admission on length of stay in hospital based on the negative binomial model which includes type of admission as covariate but not age group.

SOLUTIONS

Urgent admissions were associated with stays that were on average 1.27 (95% CI 1.15 to 1.40) times longer than those resulting from elective admissions, while emergency admissions were associated with stays that were on average 2.07 (95% CI 1.78 to 2.40) times as long.

*Note that if we wanted an overall p-value testing the null that length of stay varied by type of admission, we could perform a likelihood ratio test of the model with **type2** and **type3** as covariates with the null model (only an intercept).*

Question 8

Use the estimates from your negative binomial model that includes only type of admission to calculate the estimated means and variances for each type of admission. Note that Stata uses an different parametrisation of the model than R. In the parametrisation used by Stata, if the expectation of the outcome is μ then its variance is $\mu(1 + \alpha\mu)$.

Then compare these with the sample means and SDs for `los` in the three admission types. What do you conclude about the model?

SOLUTIONS

From the fitted model the predicted mean lengths of admission are $e^{2.1782} = 8.83$ days, $e^{2.1782+0.2373} = 11.20$ days and $e^{2.1782+0.7253} = 18.24$ days for elective, urgent and emergency admissions respectively. These are identical to the observed means in the three groups.

Since the estimate of α is 0.4478 the estimated variances are:

- $8.83 \times (1 + 0.4478 \times 8.83) = 43.7 \text{ days}^2$,
- $11.20 \times (1 + 0.4478 \times 11.20) = 67.3 \text{ days}^2$ and
- $18.24 \times (1 + 0.4478 \times 18.24) = 167.2 \text{ days}^2$

for elective, urgent and emergency admissions respectively.

These can be compared with the observed variances (use `summ los if type1==1, detail` etc.) which are 41.7, 77.9 and 424.9 days² respectively.

The sample variances are close to those predicted by the negative binomial model for the elective and urgent admissions. However, for the emergency admissions the sample variance is quite a bit larger than predicted by the model. The number of emergency admissions is relatively small, meaning that the variance will be imprecisely estimated but the lack of agreement between the fitted and observed variances could also indicate that the Gamma random-effects distribution is not appropriate for these data (or that a different is needed for each type of admission).

Another issue that we have overlooked in our analyses is that the hospital days only take positive values, whereas Poisson distributions (and negative binomials) should also sometimes take the value zero. One approach which could be used to allow for this is to instead model “number of days of stay after day of admission”, which would take the value zero for some patients.